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SNIP, A STATISTICS-SENSITIVE BACKGROUND TREATMENT FOR THE QUANTITATIVE ANALYSIS OF PIXE SPECTRA IN GEOSCIENCE APPLICATIONS

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Statistical fluctuations in X-ray spectra must be treated properly for reliable quantitative PIXE analysis. A background approximation that provides reliable treatment of fluctuations, the Statistics-sensitive Non-linear Iterative Peak-clipping (SNIP) algorithm, is described. Monte Carlo simulation demonstrates the stability of this background approximation, and hence the deduced trace element concentrations, over a wide range of counting statistics.

1. Introduction

X-ray spectra obtained in geoscience applications of a proton microprobe utilizing proton induced X-ray emission (PIXE) place particular demands on spectrum analysis and quantification due to the variety of matrix compositions and the large range of peak heights in such spectra. Consequently, the accuracy and reliability of the the trace element analysis depend critically on the treatment of the spectrum in order to resolve strong line overlaps, to account for continuum background contributions, and to distinguish artefacts related to the response of the Si(Li) detector. In particular, the treatment of background must satisfy two requirements. Firstly, a background approximation must pay particular attention to the reliable estimation of the background continuum under low-statistics peaks, both isolated and in close proximity to strong matrix lines (eg. ~ 10 ppm Ni in the presence of ~ 10 wt.% Fe). Secondly, in order to process data from numerous spot analyses efficiently and reproducibly, a background approximation must be free of user-adjustable parameters to permit batch processing.

Throughout this paper the term "low-statistics" is used to indicate a low number of accumulated counts per channel, which is characterized by pronounced statistical fluctuations, and conversely "high-statistics" refers to high counts and insignificant statistical fluctuations.

A number of approaches to background treatment have been reported [1] using polynomial fits [2], approximate physical models for the background, Fourier transform techniques [3], digital filters, families of parabolas [4], iterative line stripping [5], peak clipping [6] or filing algorithms [2] and the treatment of first and second derivative spectra [7]. A commonly adopted approximation is to reduce the statistical fluctuations in the spectrum by applying a smoothing operation, often in the form of n-point means, prior to the application of the background approximation. Such a procedure inevitably suffers from two conflicting tendencies: (1) In the high-statistics portions of the spectrum, where smoothing is really unnecessary, minima between X-ray peaks are raised, making the reliable estimation of the background even more difficult. (2) In low-statistics regions of the spectrum the smoothing is often too mild, constrained by the desire to minimize the effects of (1), resulting in significant positive and negative statistical excursions of the channel count. Such residual fluctuations can seriously confuse most simple background algorithms, which are otherwise attractive due to their robustness. As a consequence strong peak areas may be underestimated while the areas of weak, low statistics peaks may be overestimated and very variable.

The latter effect is of particular concern to reliable PIXE trace element analysis, because trace element peaks are generally weak and of low statistics. It leads to a progressive overestimate of the lower trace element concentrations. In addition, significant changes in integrated beam charge between runs on otherwise similar mineral grains produce spectra of varying statistics although representing similar trace element concentrations. As a consequence, systematic errors in the deduced concentrations can be introduced due simply to variations in the choice of a suitable integrated charge.

Hence, accurate PIXE trace element microanalysis in the geosciences demands a reliable treatment of the low-statistics portions of the spectrum (as the first phase of a background algorithm) that leaves the high-statistics portions of the spectrum unaffected. In addition,

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the background algorithm must be independent of the energy calibration of the detection system, free of the need for user-adjustable parameters, rapid and able to treat spectra of arbitrary complexity. This paper describes such an algorithm, now in routine use for the processing of proton microprobe PIXE spectra at the CSIRO Heavy Ion Analytical Facility (HIAF) [8].

2. Low statistics digital filter

The Low Statistics Digital Filter described here was developed to provide n-point mean smoothing of the data, primarily in areas of low statistics, while avoiding spreading of the bases of peaks and the degrading of minima between peaks. An adaptive n-point smooth is applied selectively to the low-statistics regions of the spectrum between peaks. The width of the interval is graded to decrease as the channel counts increase. In addition, the slope across the smoothing interval is used to detect the onset of a peak, and causes a reduction of the interval to avoid the incorporation of the peak tail in the average.

The algorithm processes a spectrum, channel by channel, in a single pass. For each channel x in the spectrum with counts $y(x)$ two adjacent windows, one on each side of the channel and $f \cdot F(E)$ channels wide, are summed producing L (left sum) and R (right sum). $F(E)$, in channel units, is the characteristic full-width at half-maximum (FWHM) of Si(Li) X-ray peaks at the energy $E(x)$ of the channel. Both windows are reduced in width until the total sum $S = y + L + R$ falls below a minimum M, which sets the base degree of smoothing in regions of vanishing counts, or until two conditions are met:

(1) The sum S is less than a cut-off $N = A\sqrt{y}$, where A is a constant. Since the ratio N/y reduces as $1/\sqrt{y}$ with increasing counts, this condition ensures that smoothing is confined to the low-statistics regions of the spectrum, and grades smoothly with channel count.

(2) The slope $(R + 1)/(L + 1)$ is confined to be within the range $1/r$ to r, where r is a constant, thus avoiding the incorporation of the sides of peaks in the average.

With these conditions satisfied, the average S/W is adopted as the channel count, where $W = 2fF + 1$, and consideration passes to the next channel. The following parameter values have been found to be successful in treating all spectra encountered to date: $f = 1.5$, $A = 75$, $M = 10$, and $r = 1.3$, and have been adopted for all PIXE microprobe work in our laboratory.

An illustration of the performance of the Low Statistics Digital Filter is shown in fig. 1a for the case of a PIXE microprobe spectrum from a spot analysis of a single mineral grain. For comparison, the effect of a simple 7-point mean smooth on the same spectrum is also shown (fig. 1b). The 7-point smooth provides barely adequate smoothing, in this case, in regions of low counts, but tends to fill the valleys between peaks. In particular, the valley between the Ni and Cu K-lines changes dramatically, making the background estimation uncertain. Hence a 7-point smooth would lead to significant variability in the determination of the Ni concentration. Clearly, it is difficult to find an adequate compromise between low- and high-statistics regions of

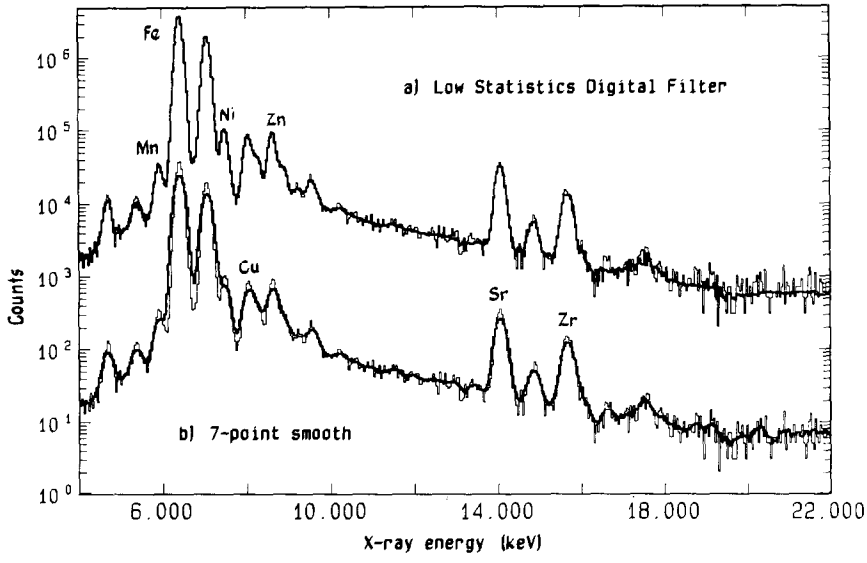


Fig. 1. PIXE spectrum from a mineral grain (augite), processed with a) the Low Statistics Digital Filter (offset ×100 for clarity), and b) a spectrum smoothing algorithm using 7-point means.

the spectrum when using a simple n-point smooth. The low statistics filter, on the other hand, attenuates background fluctuations between peaks without smearing either the peaks or the valleys between them.

3. The background algorithm

The peak filing algorithm [2], a variation of the peak clipping technique [6], formed the starting point for the development of a background algorithm to process the results of the low statistics digital filter. A significant improvement in algorithm efficiency (fewer iterations required) came from the use of a broad sampling interval, which scaled with the FWHM characteristic of the X-ray energy of the channel in the spectrum. Only ~ 100 passes were then required, compared to ~ 700 for the peak filing approach. The average of the counts at each end of the sampling interval was compared with the channel count at the centre of the interval and the lesser of the two values accepted as the new channel count. The process was repeated for every channel. A further advantage of this approach was that the peak clipping also became independent of the energy calibration of the spectrum.

A remaining problem is the large range of the counts in spectra, ranging from zero to $\sim 10^7$. The mean of two values, one large (e.g. $\sim 10^6$) and one small (e.g. ~ 10) is itself large (e.g. $\sim 5 \times 10^5$). The compression of dynamic range by using $\log_e$(counts), reduces the mean used in each comparison (e.g. $\sim 3 \times 10^3$). As a consequence peak clipping proceeds more rapidly, requiring fewer iterations. The application of a second $\log_e$ transformation compresses the range 3–10^7 counts on to the interval 0–2.8, and hence provides a further improvement to peak clipping efficiency, with a comcomitant reduction in the iteration count (typically $\sim$ 20). Further log transformations offer little benefit and may introduce rounding error problems.

Incorporating these ideas, the final Statistics-sensitive Non-linear Iterative Peak-clipping (SNIP) algorithm is structured into three phases. The first phase applies the low statistics digital filter, and transforms the channel count $y(x)$ in channel x according to $z = G(y)$, where

$$G(y) = \log\{\log(y+1)+1\},$$

in order to compress the dynamic range of the channel counts. The second phase is a multipass peak clipping loop which replaces each channel value $z(x)$ with the lesser of $z(x)$ and the mean $\bar{z}[x, w(x)]$, where

$$\bar{z}(x, w) = [z(x+w) + z(x-w)]/2$$

and the scanning width $w(x)$ is fixed to twice the X-ray peak FWHM characteristic of channel x. This approach is independent of detector energy calibration. Only 24 passes through this loop are necessary to produce acceptable background spectra for all our test cases. For the final few passes (typically 8) the scanning width is progressively reduced by $\sqrt{2}$ after each pass. This is done to reduce the incidence of oscillations in the background, and eliminates the need to smooth the background afterwards. The final phase transforms the results back to channel counts, $y = G^{-1}(z)$.

4. Tests of accuracy

As a first step in the detailed evaluation of the accuracy and reliability of the SNIP background algorithm, standard materials were analyzed and the results were compared with the nominal values. A single pass analysis scheme is used for microprobe PIXE data at HIAF for mineralogical PIXE microanalysis. Yields for all anticipated X-ray lines are calculated in "generic" form for a known major element sample composition, independent of experimental parameters and including secondary X-ray fluorescence effects in the thick target integration, using the program LAYER. Any complex layered target can be treated [9]. The relative intensities of all lines are generated, and utilized in subsequent least-squares fits of all spectra for which the matrix is appropriate. The fit is performed using the program PIXE_FIT, which incorporates the SNIP background algorithm together with modified weights to optimize peak area determination, and is implemented in vector form to utilize an FPS AP-120B array processor on a VAX 11/780 [10]. A third interactive program GEO_TRACE integrates the peak area and yield calculation results to produce trace element concentrations, collating data from a series of measurements, and facilitates the graphical display of trace element concentrations and the examination of element correlations and trends in the complete data set [10].

Glasses were prepared from the USGS rock standards AGV-1, BCR-1 and GSP-1 by melting in an Ar atmosphere on a molybdenum strip heater, and analyzed with the programs outlined above. The results, which are reported elsewhere [10,11], yielded an average deviation from the nominal values of $< 10\%$. The high Mo content is an artefact of the sample preparation [11]. The results for AGV-1 are shown in table 1. The Ni K_α peak is adjacent to the large Fe K_β peak (~ 5 wt% Fe), making the determination of the Ni peak area very independent on the background treatment. Good agreement for Ni (at ~ 15 ppm) is therefore a verification of the quality of the background algorithm, and the analysis programs, at least for this spectrum. At higher X-ray energies (above 20 keV) statistics become increasingly degraded. The good agreement for Ba results largely from the well-behaved nature of the background algorithm in regions of low statistics, together with use

Table 1
Analysis of AGV-1 [a]

	Present work [b]	Accepted values [a]
Mn	822 ±10	775
Fe (%)	(4.7)	4.7
Ni	14.2± 1.2	16
Cu	59.5± 1.2	60
Zn	91.7± 2.3	86
Ga	21.4± 1.0	20.5
Pb	36.4± 2.1	35
Th	5.6± 1.6	6.5
Rb	75.0± 1.8	67
Sr	703 ±13	660
Y	20.4± 1.4	20.5
Zr	241 ± 9	225
Nb	13.4± 1.8	15
Ba	1146 ±50 [c]	1210

[a] See refs. [10,11].
[b] Analyses normalized to the accepted Fe value, 1σ uncertainty estimates are given.
[c] Some inhomogeneity in Ba was observed [11].

of the modified least-squares fitting procedure and the inclusion of seven Ba K-lines in the fit [10].

5. Test of reliability

A complementary approach was developed to consider the question of reliability of the background algorithm under the effects of changing statistics. The statistics or intensity of a spectrum can be varied simply by the choice of different experimental conditions such as integrated beam charge. Alternatively, different intensities will characterize the analysis of different minerals. In either case, it is desirable to place on a background treatment the requirement that the performance observed using a suite of typical test spectra can be expected for any future spectrum. This is difficult, if not impossible, to demonstrate in general. However, a powerful test of the reliability of a background treatment under the effects of changes in statistics can be made by applying Monte Carlo techniques to simulate spectra of arbitrary intensity, corresponding to arbitrary integrated beam charge. Employing a high-statistics spectrum as the parent probability distribution, low-statistics spectra can be generated and processed independently with the background algorithm and fitted to deduce trace element concentrations. These concentrations can then be compared to an analysis of the parent spectrum in order to detect systematic error trends.

The Monte Carlo simulation is straightforward. The high-statistics parent spectrum $y(x)$ is first smoothed using the Low Statistics Digital Filter and then integrated, channel by channel, to form $h(x) = \int_0^x y(t)\,dt$. This spectrum is normalized and used to map a random number generator on to the full channel range. Thus for each simulated count, the channel or peak height x is given by $x = g^{-1}$ (ran), where $g(x) = h(x)/h(x_{max})$, and the random numbers "ran" are generated in the interval (0, 1).

An AGV-1 spectrum, representing an integrated beam charge of $Q = 14$ μC (fig. 2), was selected as the parent spectrum in order to generate low-statistics test spectra using the Monte Carlo approach (fig. 2). Ten

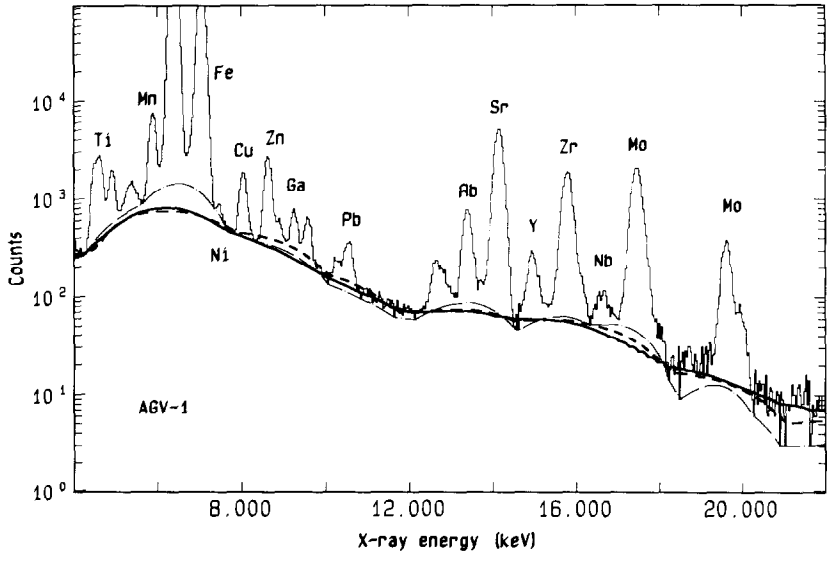


Fig. 2. PIXE spectrum from the AGV-1 glass (3 MeV protons, $Q = 14$ μC) showing the SNIP background (thick solid curve), the PC background (dot–dash curve), and the FP background (dashed curve).

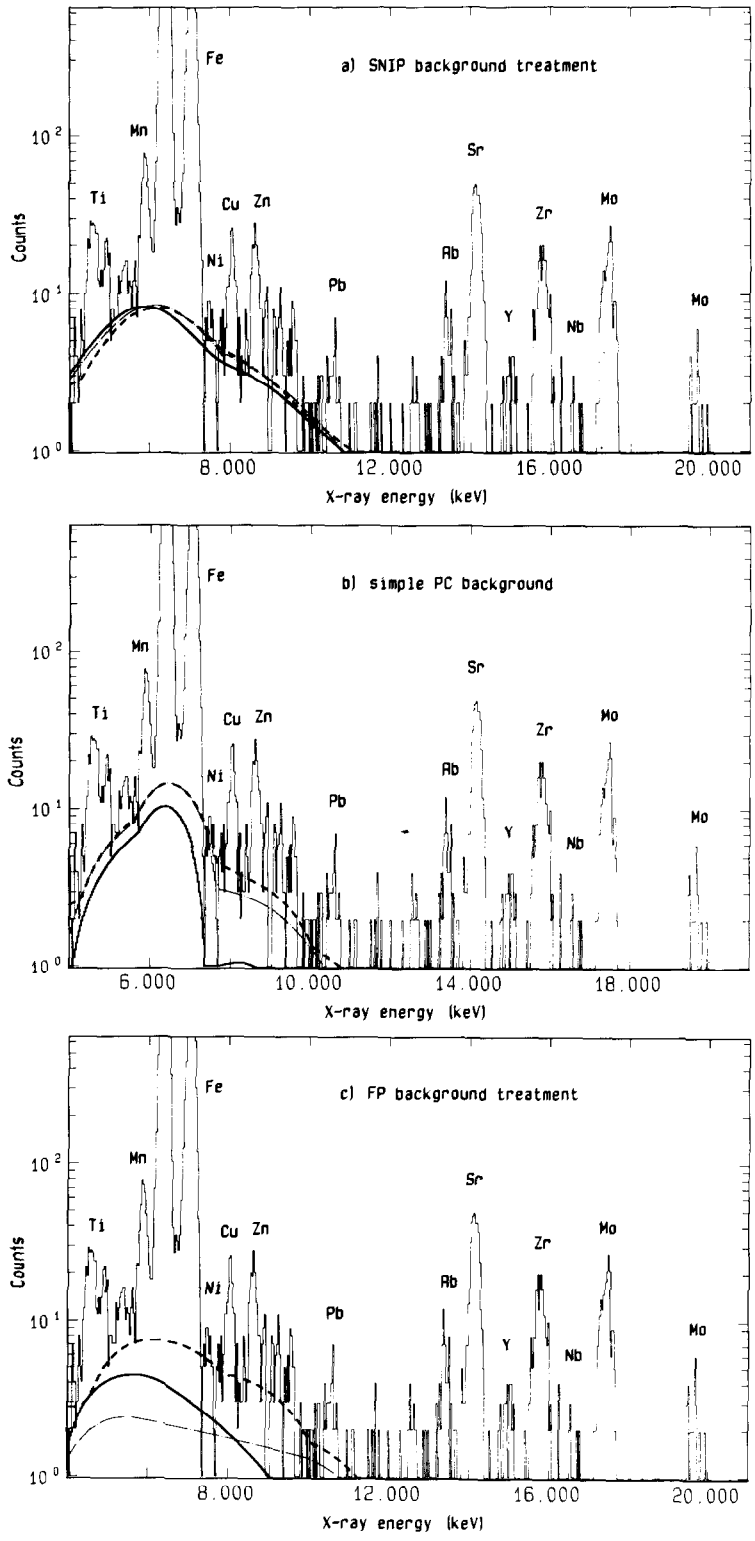


Fig. 3. A typical simulated AGV-1 spectrum representing $Q = 0.14$ μC showing as thick solid curves the a) SNIP background, b) PC background, and c) FP background approximations. Also shown are the backgrounds calculated for a $Q = 1.4$ μC spectrum (dot–dash curves, scaled by 0.1) and the $Q = 14$ μC spectrum (dashed curves, scaled by 0.01).

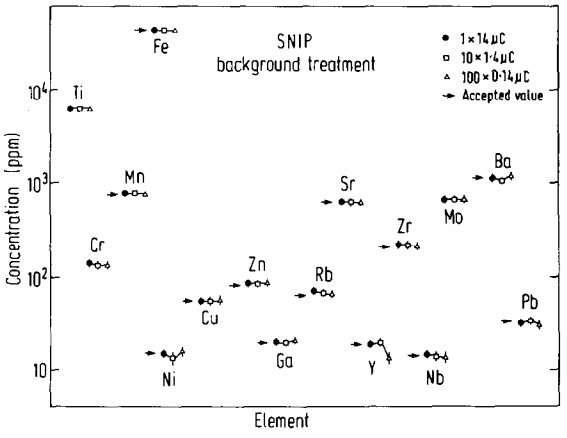


Fig. 4. Concentrations deduced from the parent AGV-1 spectrum using the SNIP background treatment (solid symbols), and the averages of concentrations deduced from the simulated spectra (open squares, average of 10 spectra of $Q = 1.4$ μC each, and open triangles, average of 100 spectra of $Q = 0.14$ μC each).

spectra of 0.1 times the charge of the parent ($Q = 1.4$ μC) were generated. These were analyzed separately using the SNIP background treatment described above. Naturally, the individual results display a scatter reflecting the reduced statistics. But the arithmetic average concentrations deduced from all 10 spectra should yield concentrations consistent with the parent values, within statistics. Indeed, the averages exhibit good agreement with the concentrations deduced from the parent spectrum and the nominal values (fig. 4).

The evaluation was taken one step further with the simulation of 100 spectra of only $Q = 0.14$ μC of integrated charge. Fig. 3a shows clearly the stability of the SNIP background treatment following a factor of 100 change in statistics. The averages were again formed and despite the poor statistics of the individual spectra, show good agreement with both the higher-statistics results and the nominal values (fig. 4). In particular, the results for Ni and Nb are noteworthy as the peak areas for these elements in the $Q = 0.14$ μC spectra are well below the detection limits. The low result for Y, which is again below the detection limit, may simply be statistical or it may indicate the onset of the incorporation of undetected peaks in the smoothed background. At any rate, these results constitute compelling evidence for the insensitivity of the background algorithm to the effects of changing statistics, whether these changes reflect varying choices of integrated beam charge or a spectrum of different intensity from an unknown mineral grain under investigation.

The effect of statistical fluctuations on the analysis of these spectra was highlighted by repeating the analysis using a simple peak clipping (PC) approach. The PC algorithm used no spectrum smoothing, and compared each channel count to the mean of the immediately adjacent channel counts. A pass count of 1000 was chosen. It is evident that while a PC background provides a practical approximation to background at high counts (fig. 2), it fails in regions of low counts (fig. 3b). Application of an n-point smooth would help, but at the expense of elements such as Ni and Nb. The average concentrations deduced with the PC background for all spectra are shown in fig. 5. The deduced concentrations for all elements display a trend consistent with the overestimation of peak areas with a reduction in peak counts. This trend is particularly pronounced for the low concentration elements Ni, Y and Nb, but also for Ba which suffers from a low count rate due to a low X-ray production cross-section and a declining detector efficiency. Clearly, the determination of the concentration of a trace element near the detection limit using this algorithm could be in doubt by as much as a factor of 4.

In order to provide a more realistic comparison, the parent and simulated spectra were also analyzed using the elegant algorithm devised by Kajfosz and Kwiatek [4] which defines the background using a family of parabolas (FP). The FP algorithm provides an inherent 5-point mean smooth which results from a spectrum compression necessitated by the length of the calculation [4]. The original shape parameters ($X = 10$, $Y = 8$ [4]) were not found to be general, although corrected for channel dispersion, and are inappropriate for these spectra. Therefore, the following tests were performed using the shape parameters $X = 10$ and $Y = 3.5$ (figs. 2, 3c). The performance of the FP background algorithm is summarized in fig. 6. The concentrations deduced from the parent spectrum are in good agreement with

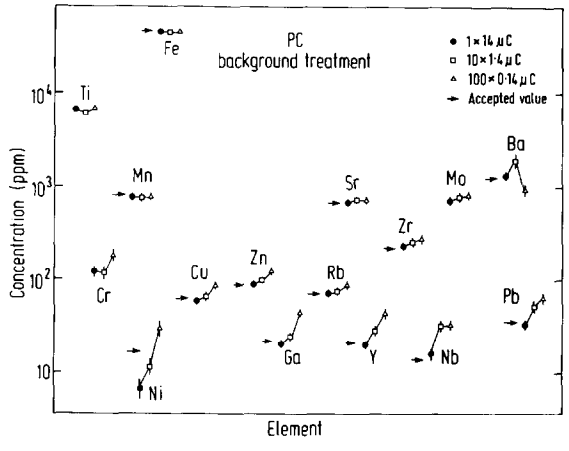


Fig. 5. Concentrations deduced from the parent AGV-1 spectrum using the PC background treatment, and the averages of concentrations deduced from the simulated spectra (symbols as in fig. 4).

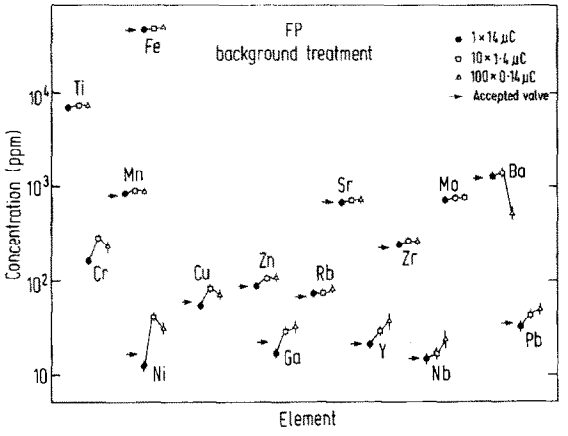


Fig. 6. Concentrations deduced from the parent AGV-1 spectrum using the FP background treatment, and the averages of concentrations deduced from the simulated spectra (symbols as in fig. 4).

the nominal values and the results of the SNIP algorithm. However, at low statistics a significant trend towards the overestimation of peak areas occurs which, as in the case of the simple PC algorithm, is symptomatic of an inappropriate treatment of statistical fluctuations. Indeed, the background displays little stability to changes in statistics (fig. 3c).

The robustness of each background algorithm against the effects of changing statistics is graphically depicted in fig. 7 as the ratio r, for each element, of the average concentration deduced from the 100 spectra, of $Q = 0.14$ μC each, to the concentration deduced from the parent spectrum. The ratios deduced using the SNIP algorithm exhibit remarkable consistency between high- and low-

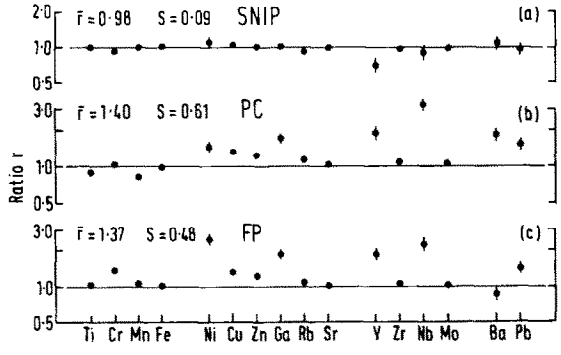


Fig. 7. The ratio r, for each element, of the average concentration deduced from the 100 spectra of $Q = 0.14$ μC each to the concentration deduced from the parent spectrum, using the a) SNIP, b) PC and c) FP background treatments.

statistics peaks (the average ratio $\bar{r} = 0.98$, and the standard deviation $s = 0.09$). In contrast, the ratios for the PC and FP algorithms depart from unity, and significantly overestimate peak areas as the statistics are reduced.

6. Conclusions

The results of the Monte Carlo simulations demonstrate that the determination of low trace element concentrations could be in serious error if statistical fluctuations are not adequately treated in the background approximation. The SNIP background algorithm provides an effective solution to this problem. The concentrations of trace elements deduced using this algorithm display remarkable stability over a wide range of counting statistics, maintaining reliable concentration values for peak areas down to the limits of detection.

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